

# HURWITZ-SPANNUNGSFELDER IM EABC-RAUM, SPEKTRALE STARRHEIT UND EIN GEOMETRISCHES RAHMENMODELL ZUR RIEMANNSCHEN VERMUTUNG

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**ABSTRACT.** We formulate a geometric–spectral framework for the Riemann Hypothesis on an adelicly inspired configuration space, called the EABC-space. The starting point is a combinatorial Laplacian defined on prime-induced configurations, together with a fourth-order Hamiltonian obtained as its square. Under explicit structural assumptions — in particular on heat-trace scaling, spectral rigidity, and the existence of stabilizing correction terms — the critical line  $\operatorname{Re}(s) = \frac{1}{2}$  emerges as the energetically preferred locus of the nontrivial zeros of the Riemann zeta function.

This article does not claim a classical proof of the Riemann Hypothesis in the strict analytic-number-theoretic sense. Rather, it develops a coherent semi-formal program in which the hypothesis appears as a consequence of an arithmetic-geometric stability principle. The notions of Hurwitz tension, symbolic curvature, arithmetic Lamb shift, and torsional correction are introduced axiomatically and related to an operator-theoretic setting. A numerical four-family spectral decomposition modulo 12 is included as a motivating computational observation.

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## 1. INTRODUCTION

The Riemann Hypothesis is among the most fundamental open problems in mathematics. A wide range of approaches — analytic, spectral, geometric, probabilistic, and physically inspired — suggest that the critical line

$$\operatorname{Re}(s) = \frac{1}{2}$$

should be interpreted not merely as a formal symmetry axis, but as a distinguished structural locus.

The aim of the present paper is to formulate a geometric framework in which prime numbers act as carriers of a spectrally rigid arithmetic medium. This medium is encoded by an adelicly inspired configuration space  $\mathcal{M}_A$  and by discrete operators acting on prime-induced combinatorial structures. The guiding principle is that any displacement of a nontrivial zero away from the critical line produces an energetic instability incompatible with global spectral coherence.

The text has a deliberately two-layered structure:

1. a formal layer consisting of definitions, operators, asymptotic laws, and propositions;
2. an interpretive layer in which these objects are viewed as curvature, tension, and stabilization mechanisms.

## 2. THE EABC-SPACE AS ARITHMETIC CONFIGURATION SPACE

**Definition 2.1** (EABC-space). We define the *EABC-space* as an abstract arithmetic configuration space  $\mathcal{M}_A$  whose local data consist of prime-based configurations, interaction laws between them, and an associated tension functional. The acronym EABC stands for *Extended Adelic Boundary Conditions*.

*Remark 2.2.* The word “space” is used heuristically. For the mathematical development below, it is sufficient to regard  $\mathcal{M}_A$  as a carrier for discrete operators, effective metrics, and energy-like functionals.

The local arithmetic structure is encoded by a function  $\Phi_H$ , called the *Hurwitz tension*.

**Definition 2.3** (Hurwitz tension). A Hurwitz tension is a scalar or tensorial quantity

$$\Phi_H : \mathcal{M}_A \rightarrow \mathbb{R}$$

measuring the local irregularity of arithmetic configurations. Its gradient is interpreted as an effective forcing term.

## 3. DISCRETE OPERATORS AND SPECTRAL DIMENSION

Let

$$P_N = \{p_1, \dots, p_N\}$$

be a finite set of primes. Suppose that a graph or hypergraph structure is imposed on  $P_N$ , with adjacency operator  $K$  and degree matrix  $D$ .

**Definition 3.1** (Combinatorial Laplacian). The unnormalized combinatorial Laplacian is

$$L_c = D - K.$$

The dynamical object of primary interest is the squared Laplacian.

**Definition 3.2** (Fourth-order Hamiltonian). The associated Hamiltonian is

$$H = L_c^2.$$

This is a positive semidefinite discrete bi-Laplacian.

Its spectral information is encoded in the heat trace

$$\Theta(t) = \operatorname{Tr}(e^{-tH}).$$

**Assumption 3.3** (Heat-trace scaling). As  $t \downarrow 0$ , the heat trace satisfies the asymptotic law

$$\Theta(t) \sim C t^{-1/4}$$

for some constant  $C > 0$ .

**Definition 3.4** (Effective spectral dimension). Whenever

$$\Theta(t) \sim C t^{-\beta},$$

we define the effective spectral dimension by

$$d_s := 2\beta.$$

Hence, under the previous assumption,

$$d_s = \frac{1}{2}.$$

*Remark 3.5.* This definition is modeled on standard heat-kernel asymptotics. In the present discrete context,  $d_s$  is to be understood as an effective scale parameter rather than as a canonical smooth dimension.

#### 4. SPECTRAL RIGIDITY AND SYMBOLIC CURVATURE

The interpretation of

$$d_s = \frac{1}{2}$$

is that of maximal spectral compression. In heuristic physical language, the arithmetic medium carries fewer diffusive degrees of freedom than an ordinary geometric background.

**Definition 4.1** (Spectral rigidity). We call the model *spectrally rigid* if admissible perturbations of the local arithmetic configuration do not alter the asymptotic heat-trace scaling exponent.

In addition, irregular prime gaps are treated as sources of symbolic curvature.

**Definition 4.2** (Symbolic curvature). The *symbolic curvature* is an effective quantity

$$\kappa : \mathcal{M}_A \rightarrow \mathbb{R}$$

whose gradient  $\nabla \kappa$  measures local irregularities in arithmetic resonance patterns.

*Remark 4.3.* This curvature is not Riemannian curvature in the differential-geometric sense. It is an effective structural quantity derived from the model.

#### 5. CORRECTION TERMS: ARITHMETIC LAMB SHIFT AND TORSION

The central stabilizing mechanism of the framework is expressed through two correction terms.

**Definition 5.1** (Arithmetic Lamb shift). The *arithmetic Lamb shift* is a first relevant stabilizing contribution of the form

$$\Delta E_{\text{stab}} = \alpha^5 \nabla \kappa,$$

where  $\alpha$  is a dimensionless coupling parameter.

**Definition 5.2** (Arithmetic torsion). The *arithmetic torsion* is a higher-order correction term, symbolically of size  $\alpha^6$ , controlling collective feedback between resonance clusters.

**Assumption 5.3** (Stabilization principle). In the stable regime, the total correction mechanism compensates all admissible forcing terms generated by  $\nabla \kappa$ .

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FIGURE 1. Empirical spectral decomposition of primes up to  $N = 200000$  into the four nontrivial residue classes modulo 12:  $E \equiv 1 \pmod{12}$ ,  $A \equiv 5 \pmod{12}$ ,  $B \equiv 7 \pmod{12}$ ,  $C \equiv 11 \pmod{12}$ . Shown are corrected amplitudes as functions of normalized frequency. The dominant spectral peaks appear at closely aligned frequency locations across all four families. In the  $A$ -family, a local damping at  $p = 137$  is marked explicitly; within the model this is interpreted heuristically as a test instance of an arithmetic Lamb-shift-type correction. The figure is intended as numerical motivation rather than as an independent proof component.

## 6. THE CRITICAL LINE AS ENERGETIC MINIMUM

The geometric heart of the framework is the assumption that the critical line is the global minimum of an effective energy functional.

**Assumption 6.1** (Energy minimization principle). There exists an effective functional

$$\mathcal{E}(\rho)$$

defined on the set of nontrivial zeros  $\rho$  of  $\zeta$ , such that

$$\mathcal{E}(\rho)$$

is minimized precisely when

$$\operatorname{Re}(\rho) = \frac{1}{2}.$$

*Remark 6.2.* This is the decisive structural assumption of the model. Without it, no sufficiently strong link between spectral rigidity and zero location can be derived.

## 7. A FOUR-FAMILY SPECTRAL DECOMPOSITION MODULO 12

A natural arithmetic decomposition of the odd primes is given by the four nontrivial residue classes modulo 12,

$$1, 5, 7, 11 \pmod{12}.$$

Within the present framework, we refer to these classes as the four families

$$E, A, B, C.$$

Figure 1 displays an empirical spectral decomposition of these four families for primes up to  $N = 200000$ . The horizontal axis represents normalized frequency; the vertical axis represents a corrected amplitude derived from the chosen computational pipeline.

The main empirical observation is that the four spectra do not behave as independent random objects. Instead, dominant peaks recur at closely matching normalized frequencies across all four families. In particular, visually prominent lines occur repeatedly near frequency bands around

$$f \approx 0.08, 0.17, 0.25, 0.33, 0.42, 0.50,$$

although the precise values depend on normalization, preprocessing, and the chosen spectral transform.

At least three different interpretations are possible:

1. the pattern may be an artifact of the common embedding of all four families into the global prime density;
2. it may reflect a deeper modular coherence among residue classes modulo 12;
3. it may indicate that the primes, viewed spectrally, organize into a coupled multi-family resonance structure rather than into independent residue-class components.

Within the EABC framework, the third interpretation is provisionally favored: the four families are regarded as projections of a common arithmetic resonance system. The observed peak alignment is therefore read as evidence — at least at a heuristic level — for a form of spectral rigidity extending across residue classes.

At the same time, caution is necessary. Shared peak locations alone do not imply a rigorous theorem. To extract a mathematically robust conclusion from this figure, one would need at least:

- a) an exact definition of the spectral transform used;
- b) control of windowing, normalization, and smoothing effects;
- c) a null model based on randomized or surrogate prime sequences;
- d) a stability analysis of the dominant peaks under variation of the cut-off  $N$ .

**7.1. Local correction at  $p = 137$ .** In the  $A$ -family, Figure 1 marks an explicit local damping at  $p = 137$ . In the language of the model, this modification is called an *arithmetic Lamb-shift correction*.

Formally, however, the terminology should at present be understood only heuristically. What is actually implemented is a localized renormalizing perturbation of a single arithmetic contribution, introduced in order to test the sensitivity of the global spectral picture. The relevant numerical question is therefore not whether 137 must be intrinsically distinguished, but whether small local interventions at arithmetically prominent sites induce disproportionate changes in the global spectrum.

If such sensitivity can be reproduced systematically, then certain primes may act as resonance-relevant nodes within the model. Without a broader sensitivity analysis, however, this interpretation remains provisional.

## 8. MAIN STRUCTURAL PROPOSITION

We now formulate the logical core of the framework.

**Proposition 8.1.** *Assume that:*

- (i) *the EABC-space  $\mathcal{M}_A$  is singularity-free in the sense of the model;*
- (ii) *the Hamiltonian  $H = L_c^2$  satisfies*

$$\Theta(t) \sim C t^{-1/4};$$

- (iii) *this implies the effective spectral dimension*

$$d_s = \frac{1}{2};$$

- (iv) *any displacement of a nontrivial zero away from the critical line induces a nontrivial forcing term via  $\nabla \kappa$ ;*
- (v) *this forcing term can be globally compensated if and only if the zero lies on the critical line.*

*Then the critical line*

$$\operatorname{Re}(s) = \frac{1}{2}$$

*is the unique globally stable locus for nontrivial zeros of the Riemann zeta function.*

*Proof.* By assumptions (ii) and (iii), the system operates in a regime of fixed effective spectral dimension. This excludes macroscopic spectral deformations. By assumption (iv), any displacement of a zero away from the critical line generates an additional forcing term. By assumption (v), full compensation of this forcing term is possible only on the critical line. Therefore every configuration with

$$\operatorname{Re}(\rho) \neq \frac{1}{2}$$

is energetically unstable. Hence the only globally stable locus is

$$\operatorname{Re}(\rho) = \frac{1}{2}.$$

□

## 9. WHAT IS ESTABLISHED HERE — AND WHAT IS NOT

*Remark 9.1* (Methodological status). The previous proposition is not a proof of the Riemann Hypothesis in the standard sense of analytic number theory. What it establishes is the following conditional statement:

1. Under explicit spectral-geometric stability assumptions, the critical line is the unique globally stable locus of nontrivial zeros.
2. The framework offers a coherent bridge between prime spectra, heat-trace asymptotics, and zero localization.
3. To turn this into a full proof, the model assumptions would need to be independently justified in a mathematically rigorous way.

At minimum, the following tasks remain:

- a) a precise construction of the graph or hypergraph structure on primes;
- b) a rigorous proof of the heat-trace asymptotic law;
- c) a mathematically controlled relation between the spectrum of  $H$  and the zeros of  $\zeta$ ;
- d) a precise analytic definition of the  $\alpha^5$  stabilization term;
- e) an actual exclusion argument for off-critical zeros.

## 10. NUMERICAL OUTLOOK

Even at the heuristic stage, the framework suggests several concrete computational tests:

1. numerical estimation of the heat-trace exponent for prime-induced graphs;
2. spectral comparison with random-matrix statistics for low-lying zeros;
3. robustness checks of the exponent  $d_s = \frac{1}{2}$  under changes in adjacency structure;
4. construction of explicit toy models for the effective functional  $\mathcal{E}$ ;
5. perturbative sensitivity tests on selected primes such as  $p = 137$ .

## 11. CONCLUSION

We have formulated an arithmetic-geometric framework in which prime numbers are treated as carriers of a spectrally rigid medium and the critical line

$$\operatorname{Re}(s) = \frac{1}{2}$$

emerges as the energetically preferred location of the nontrivial zeros of the Riemann zeta function. The current contribution does not lie in a completed proof of the Riemann Hypothesis, but in the systematic synthesis of operator theory, heat-trace asymptotics, stability principles, and arithmetic geometry into a unified conceptual program.

In this perspective, the central question is not the isolated location of a single zero, but the global structure of the arithmetic space that would force all zeros to align.

## APPENDIX A. TEMPLATE FOR A NUMERICAL APPENDIX

This appendix provides a placeholder structure for future numerical validation.

A.1. **Low-lying zeros and empirical distributions.** Let

$$\rho_n = \frac{1}{2} + i\gamma_n$$

denote the first nontrivial zeros under the working hypothesis of critical-line localization. One may compare the normalized spacings

$$s_n = \frac{\gamma_{n+1} - \gamma_n}{\langle \gamma_{n+1} - \gamma_n \rangle}$$

with GUE-type predictions.

A.2. **Kolmogorov–Smirnov statistics.** If a numerical dataset is available, one may report a KS statistic of the form

$$D_{\text{KS}} = \sup_x |F_N(x) - F_{\text{ref}}(x)|.$$

A placeholder sentence could read:

For the dataset extracted from `zeros6.npy`, the empirical KS distance to the chosen reference law is reported as

$$D_{\text{KS}} = 0.009,$$

subject to the precise normalization and reference distribution specified in the accompanying computational notebook.

A.3. **Heat-trace fits.** Given eigenvalues  $\lambda_1, \dots, \lambda_M$  of  $H = L_c^2$ , one computes

$$\Theta_M(t) = \sum_{j=1}^M e^{-t\lambda_j}.$$

A log-log regression may then be used to estimate the exponent in

$$\Theta_M(t) \approx Ct^{-\beta}.$$

A.4. **Figures.** Examples of figures to include:

1. log-log plot of  $\Theta_M(t)$  versus  $t$ ;
2. histogram of normalized zero spacings;
3. spectral density plot of  $H$ ;
4. stability plot under perturbation of graph parameters.

## APPENDIX B. NOTES ON REFERENCES AND CLAIMS

If this manuscript is intended for broader circulation, it is advisable to distinguish carefully between:

1. established mathematical results,
2. numerical observations,
3. model assumptions,
4. speculative interpretations.

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